



Noren Spring

Spring Manufacturing Experts

ABOUT

APEX Group is based in Xuzhou, China, specializing in designing, developing and manufacturing high performance, quality-driven springs with about **25 years of experience**.

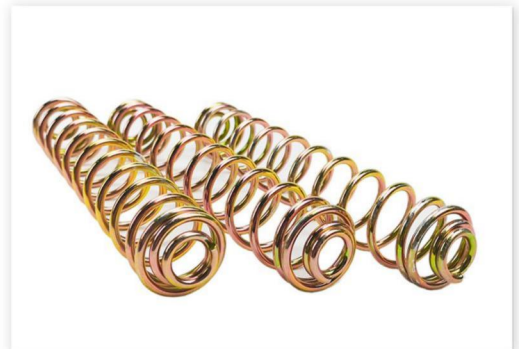
Our factory operates a **12,000 m² production base** with over **120 employees**, including a 15-engineer R&D team, and offers full in-house manufacturing for wire diameters from **0.1 mm to 80 mm**, with a monthly production capacity exceeding **4.16 million spring** products. With strong design, development, and parameter analysis capabilities, we provide customized, high-precision spring solutions for automotive, industrial equipment, and precision engineering applications. Certified to international standards including **IATF 16949, ISO 13485, ISO 9001, ISO 14001, and ISO 45001**, Our products are exported to over **60 countries and regions worldwide, serving more than 2,000 global customers**.

PRODUCTS

Compression Springs

Compression springs are designed to resist compressive forces and store energy when pushed. When the load is removed, they return to their original shape. They are widely used in automotive, machinery, and electronic devices.

Applications: automotive suspension systems, mechanical valves, shock absorbers, household appliances, electronic devices.



Extension Springs

Extension springs work by resisting pulling forces. They are tightly coiled and designed to extend under load, returning to their original length when the force is released.

Applications: garage doors, agricultural machinery, trampolines, industrial equipment, mechanical linkages.



Torsion Springs

Torsion springs operate by twisting around their axis. They store rotational energy and release torque when needed, commonly used in hinges and mechanical systems.

Applications: door hinges, clips, electrical switches, automotive parts, folding mechanisms.



Conical Springs

Conical springs have a cone shape that allows them to compress into a near-flat state. They provide stability and reduce buckling in compact spaces.

Applications: battery contacts, precision instruments, compact mechanical assemblies, automotive components.



Coil Length Springs

Coil length springs are extended helical springs designed for applications requiring long stroke or large deformation capacity while maintaining stable force characteristics.

Applications: industrial machinery, braking systems, cable systems, heavy-duty mechanical structures.



Die Springs

Die springs are heavy-duty compression springs used in stamping dies and industrial tooling. They provide high force in compact spaces and withstand repeated heavy loads.

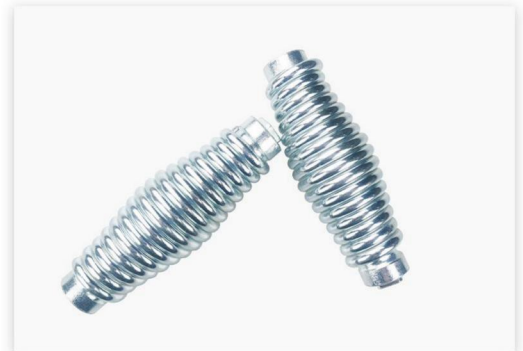
Applications: stamping dies, mold systems, press machines, industrial tooling equipment.



Vibrator Springs

Vibrator springs are designed for vibration equipment, providing elasticity and stability under continuous dynamic loads. They help reduce noise and absorb shock.

Applications: vibrating screens, feeders, construction machinery, industrial separation equipment.



Disc Springs

Disc springs are conical-shaped washers that provide high force in a very small space. They can be stacked to achieve different load and deflection requirements.

Applications: bolted joints, valve systems, clutches, heavy machinery, railway equipment.



Wave Springs

Wave springs are made of flat wire formed into waves. They replace traditional coil springs in compact spaces and reduce assembly height while maintaining performance.

Applications: motors, pumps, bearings, medical devices, precision instruments.





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